OpenMV® Thermopile Shield

Datasheet

SKU: OMV-SHIELD-THERMOPILE-V1

Description

The Thermopile Shield gives your OpenMV Cam the ability to see heat. It features a 16x4 pixel sensor array with a 60\u00B0 x 16\u00B0 field of view, capable of measuring object temperatures between -50\u00B0C and 300\u00B0C while measuring ambient room temperature (-40\u00B0C to 85\u00B0C). Your OpenMV Cam's firmware has built-in support for controlling the Thermopile Shield using the fir module.

Target Areas

Thermal imaging, temperature measurement, heat detection, HVAC diagnostics, industrial inspection, presence detection.

Key Features

- **16x4 thermal array** — 16x4 pixel thermopile sensor array.
- **60° x 16° FOV** — 60-degree horizontal by 16-degree vertical field of view.
- **-50°C to 300°C object range** — Measures object temperatures from -50°C to 300°C.
- **Ambient sensing** — Room temperature measurement from -40°C to 85°C.
- **Built-in firmware support** — Controlled via the fir module in MicroPython.

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1 Specifications

1.1 Thermal Sensor

Component	Details
Array	16x4 pixel thermopile sensor array.
Field of View	60° x 16° (horizontal x vertical).
Object Temperature	-50°C to 300°C.
Ambient Temperature	-40°C to 85°C.

1.2 Software

Component	Details
API	Built-in fir module in MicroPython firmware.

1.3 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models.

2 General Info

Dimension	Value
Weight	7 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/thermopile-shield
FIR Module Docs	http://docs.openmv.io/library/omv.fir.html
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.